

# Phase 5: Project Development Phase

SmartLenderAI-Powered Credit Card Approval System

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Team ID : SWTID-2026-4551

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## Implementation Overview

Development followed the design laid out in Phase 4. Data cleaning, feature engineering, and model training were carried out in a Jupyter notebook. The trained model, scaler, and encoders were serialized and loaded into a Flask application that exposes a form-based prediction endpoint.

## Key Modules

- **Training/** – data preparation, EDA, model training and tuning notebook
- **Flask/app.py** – request handling and prediction endpoint
- **Flask/templates/index.html** – applicant input form
- **Flask/model/** – serialized model, scaler, and encoders

## Model Results

Logistic Regression, Decision Tree, Random Forest, and XGBoost were trained and compared. XGBoost was tuned further with RandomizedSearchCV and selected as the production model based on its cross-validated accuracy.

| Model               | Accuracy |
|---------------------|----------|
| Logistic Regression | 0.762    |
| Decision Tree       | 0.741    |

|                                   |              |
|-----------------------------------|--------------|
| Random Forest                     | 0.799        |
| <b>XGBoost (tuned, 5-fold CV)</b> | <b>0.811</b> |

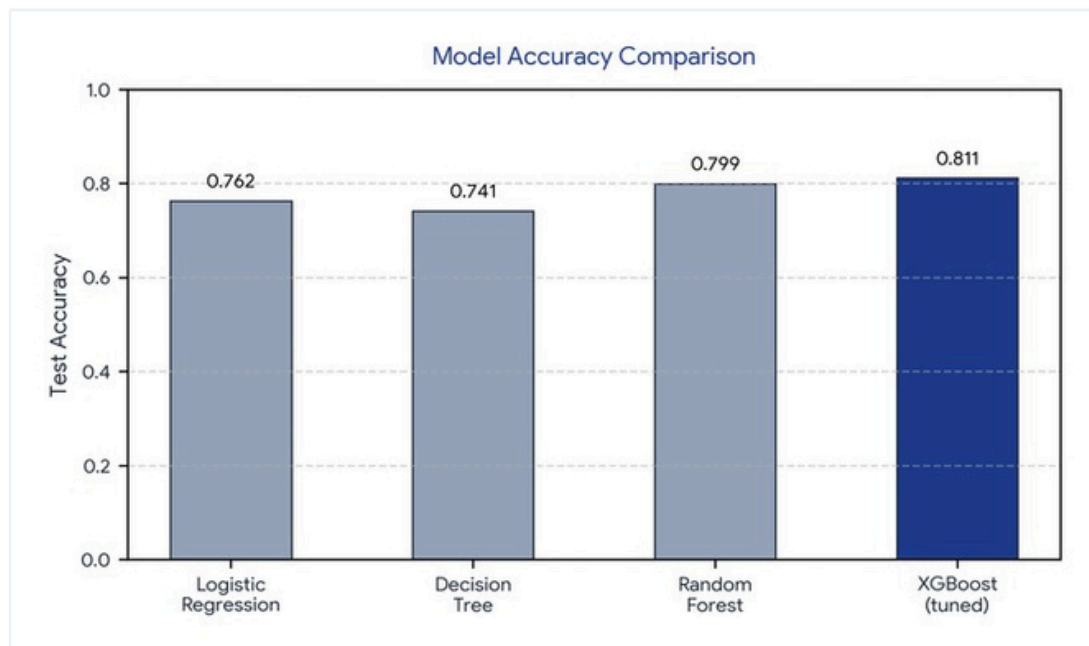


Figure 4: Model accuracy comparison